

PROJECT MILESTONE 2

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- a) A list of all implemented classes with a brief description what the purpose / function of that class is?

CANFrame	Holds the CAN data (i.e. Time Offset, Message Id, Data Length and Data Bytes)
CANSimulation	It orchestrates the parsing of CAN data File and GPS coordinates file and it also take cares for starting up HTTP server and WebSocket for GUI simulation
CANTrace	It holds the parsed CAN data and provides functionality to access CAN data on need.
CANTraceParser	Used to parse CAN data from the file given by the user
GPSCoordinate	Used to store GPS coordinates
GPSParser	Used to parse GPS data from the file given by the user
GPSTrace	It holds the parsed GPS data and provides functionality to access GPS data based on time offset.
HttpServerHandler	Handles the requests for the root endpoint by returning the index.html file
MultipleCANFrameData	Blueprint for storing multiple sensor data in a single CAN frame
SensorDataReceiver	Receives the sensor data from CANSimulation.startSimulation and transfers it to the websockets for broadcasting of data
SimulationGUI	Responsible for starting up HTTP server and WebSocket
SingleCANFrameData	Blueprint for storing single sensor data in a CAN frame
SocketHandler	Process the requests coming through the sockets (i.e. Making new connection, Broadcasting data)
SegmentData	This class stores all the values needed for the segment and calculates all the values outputted in the console.

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- b) A list of all group members and for each member what they worked on (e.g. which functions, classes, and similar the member worked on)

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GPSCoordinate	Index.html
GPSTrace	HttpServerHandler
SocketHandler	SensorDataReceiver
SimulationGUI	GPSParser
SocketHandler	CANSimulation
SegmentData	SegmentData

- c) A screenshot of the console output showing the information of all detected segments after the first simulation (run one simulation and use the print function to show all data that you have collected during the first test drive)

```
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Please enter the following port number inside the website when prompted: 50500
WebSocket server started successfully
| Current Time | Vehicle Speed | SteerAngle | YawRate | LatAccel | LongAccel | GPS Lat/Long |
| 8722.1 | ... | 2.0 | ... | ... | ... | 52.721639 / 13.225752 |
| 11121.8 | ... | 0.0 | ... | ... | ... | 52.721842 / 13.226593 |
| 11921.5 | ... | -2.0 | ... | ... | ... | 52.721842 / 13.226593 |
| 18220.2 | ... | -34.0 | ... | ... | ... | 52.722722 / 13.22802 |
| 18625.2 | ... | -36.0 | ... | ... | ... | 52.722722 / 13.22802 |
| 27628.4 | ... | -6.0 | -0.18 | -2.48 | 0.16 | 52.724135 / 13.227423 |
| 30523.9 | ... | ... | ... | ... | ... | 52.724358 / 13.227377 |
| 43275.5 | ... | 72.0 | ... | ... | ... | 0.0 / 0.0 |
<===== 75% EXECUTING [56s]
> :app:run
| SegStart | SegEnd | AvgSpeed | MaxVehSpeed | MinVehSpeed | CurveDir | MaxAccel | MinAccel | StraightLen | DegCurve | MaxSteerAngle |
| 52.721103/13.2235 | 52.722113/13.227414 | 75.14 | 78.8 | 0.0 | null | 6.46 | -1.58 | 287.11 | 0.0 | 4.0 |
| 52.722113/13.227414 | 52.723422/13.227892 | 73.24 | 77.3 | 0.0 | LEFT | 10.49 | -0.24 | 0.0 | -231.69 | -40.0 |
| 52.723422/13.227892 | 52.724225/13.227375 | 62.05 | 72.0 | 0.0 | null | 5.76 | -8.93 | 95.86 | 0.0 | -24.0 |
| 52.724225/13.227375 | 52.724417/13.227387 | 32.21 | 40.8 | 0.0 | RIGHT | 0.0 | -18.55 | 0.0 | -35.75 | 112.0 |
| 52.724417/13.227387 | 52.724417/13.227387 | 25.38 | 26.6 | 0.0 | null | 10.49 | -4.55 | 0.0 | 0.0 | -104.0 |
| 52.724417/13.227387 | 52.724352/13.226948 | 23.91 | 27.1 | 0.0 | LEFT | 28.16 | -0.32 | 0.0 | -871.11 | -214.0 |
<===== 75% EXECUTING [2m 37s]
> :app:run
```

Figure 1: Console Output of Segment Data

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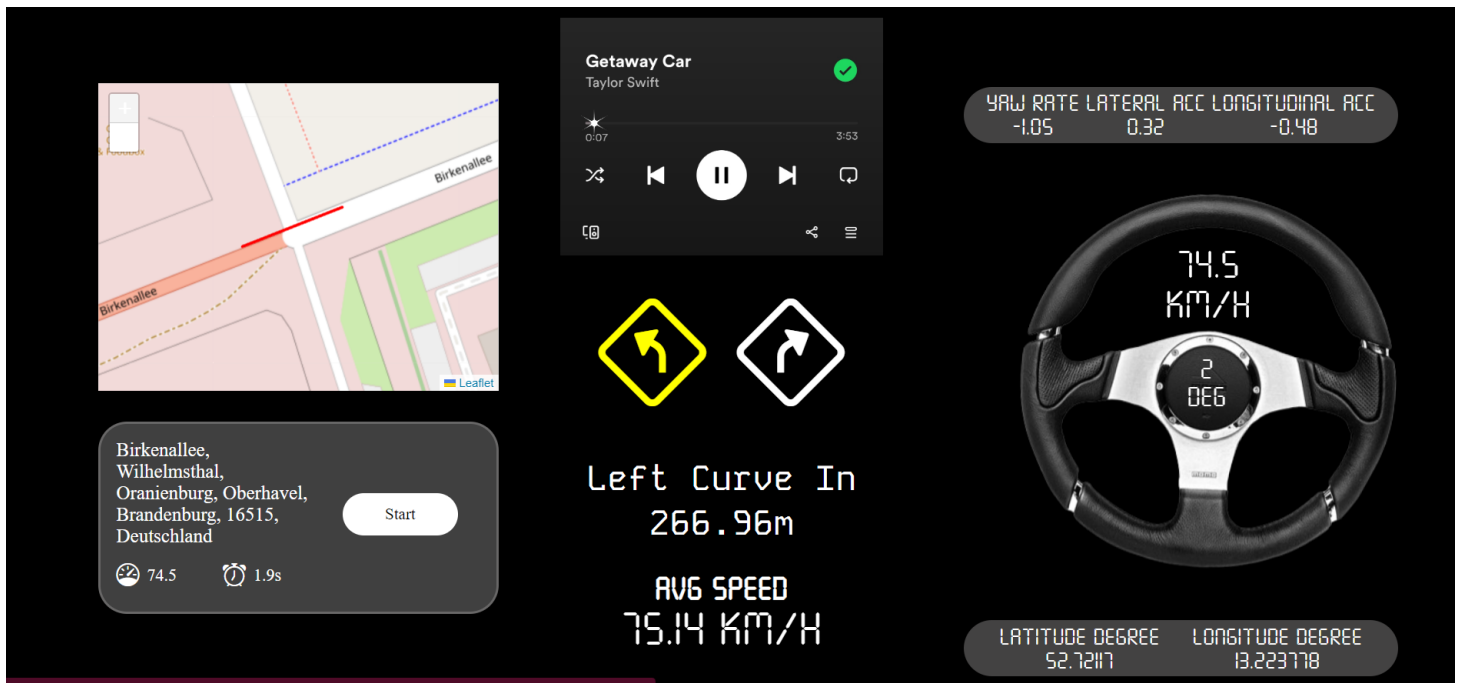


Figure 2: HMI Dashboard Indicating Curve-Ahead

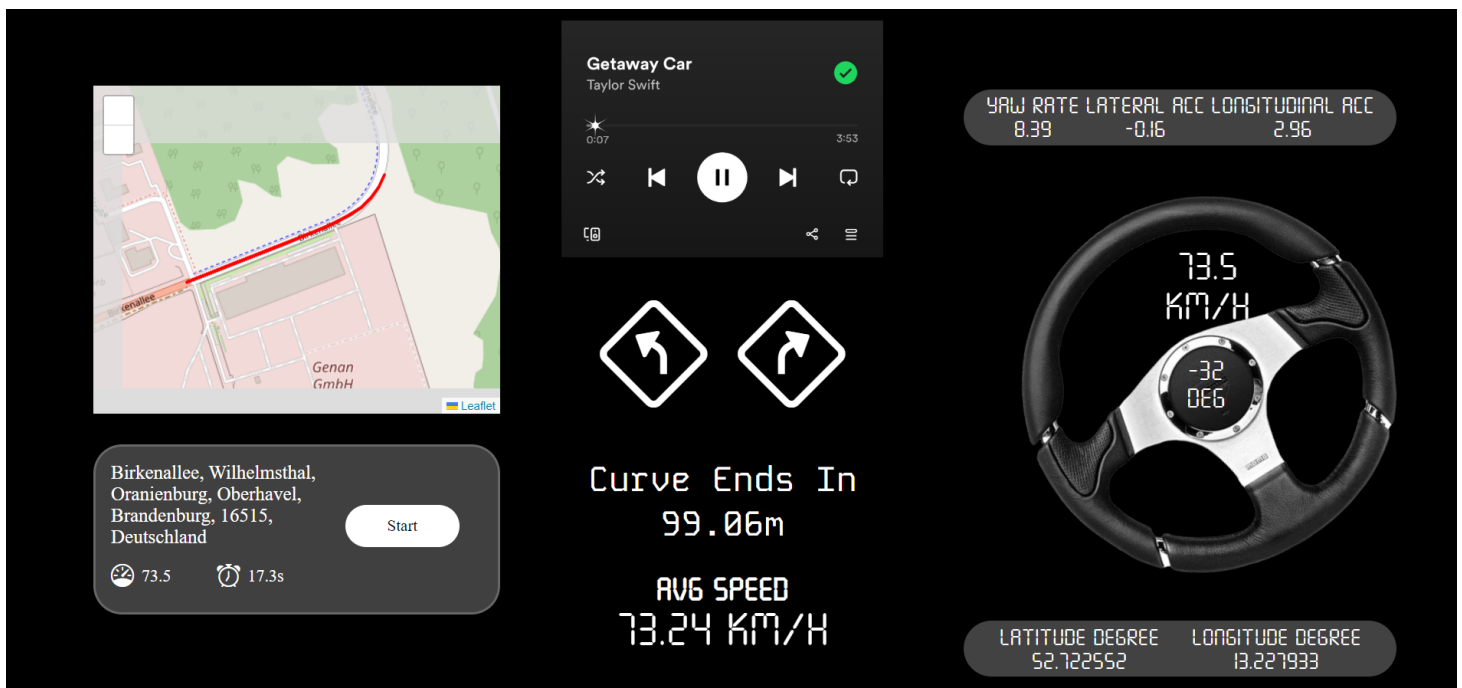


Figure 3: HMI Dashboard Indicating Curve Duration